

The Limiting Product-Measure Constants in a Question on Ramsey-Type Spheres

Run fa_banach_001, agent_lane_13

2026-06-14

Status

This is a partial solution packet for arXiv:2308.07668, “On Ramsey-type properties of the distance in nonseparable spheres” by Piotr Koszmider. It extends the earlier packet for the Hilbertian case $p = 2$. The packet proves the exact value of $c = \sup_n c_n$ for every $1 < p < \infty$, proves the exact finite values of c_n for $p \geq 2$, and gives asymptotically sharp finite bounds for $1 < p < 2$.

Source Evidence

The following crop is from page 25 of the source PDF.

and $\varphi_B \sim \varphi_{A \cap B} + \varphi_{A \cap B^c}$ as the functions characterizing A and B in the increasing order agree on $A \cap B$ as they agree with e_I . So $\|f - f'\|_{L_p}$ is not bigger than c_n by the definition of c_n , as required. $\square$

We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous. Note that $k^+(S_{L_p})$ is not $\sqrt[p]{2}$ in general because we can construct in $L_p(\{0, 1\}^{\omega_1})$ uncountable sets of independent variables assuming values 1 and -1 with equal probability $1/2$. They have norm 1 but their differences have the norm equal to $2^{1-1/p}$ which is strictly bigger than $\sqrt[p]{2}$ if p is sufficiently large.

The relevant source sentence is:

“We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous.”

Source Notation

Let μ be the product probability measure on $\{0, 1\}^\kappa$, where $\kappa \leq 2^\omega$ is uncountable. In Proposition 55 of the source paper, for $A \in [\kappa]^n$, the map T_A sends a function $F \in L_p(\{0, 1\}^n)$ to the function on $\{0, 1\}^\kappa$ depending on the coordinates in A . The constant c_n is the supremum of

$$\|T_A(F) - T_B(F)\|_{L_p}$$

over all unit vectors $F \in L_p(\{0, 1\}^n)$ and all admissible pairs $A, B \in [\kappa]^n$ whose common coordinates occur in the same positions. Finally $c = \sup_n c_n$.

Proof Intuition

Once A and B are fixed, the only probabilistic structure that matters is this: after conditioning on the coordinates shared by A and B , the two random variables $T_A(F)$ and $T_B(F)$ become iid copies of the same finite random variable. Thus the constants are governed by the sharp ratio between $\mathbb{E}|U - V|^p$ and $\mathbb{E}|U|^p$ for iid real random variables U, V . For $1 < p \leq 2$, the real line has p -negative type, giving the bound 2. For $p \geq 2$, translating to the p -center and using the monotonicity inequality for $t \mapsto |t|^{p-2}t$ gives the bound 2^{p-1} . Rare spikes attain the first bound in the limit; Rademacher variables attain the second bound at every finite level.

Lemma 1 (iid moment bound). *Let $1 < p < \infty$, and let U, V be iid real random variables with finite p -th moment. Then*

$$\mathbb{E}|U - V|^p \leq \begin{cases} 2 \mathbb{E}|U|^p, & 1 < p \leq 2, \\ 2^{p-1} \mathbb{E}|U|^p, & 2 \leq p < \infty. \end{cases}$$

Both constants are sharp in the following sense: the first is approached by rare spikes, and the second is attained by a symmetric two-point Rademacher variable.

Proof. For $1 < p \leq 2$, use the standard negative-type representation

$$|t|^p = C_p \int_0^\infty (1 - \cos(st)) s^{-1-p} ds, \quad C_p > 0.$$

Hence the kernel $|x - y|^p$ is conditionally negative definite on $\mathbb{R}$. Applying this to the signed measure $\nu = \mathcal{L}(U) - \delta_0$, which has total mass zero, gives

$$0 \geq \iint |x - y|^p d\nu(x) d\nu(y) = \mathbb{E}|U - V|^p - 2\mathbb{E}|U|^p.$$

Now suppose $p \geq 2$, and write $\phi(t) = |t|^{p-2}t$. For all real x, y ,

$$|x - y|^p \leq 2^{p-2}(x - y)(\phi(x) - \phi(y)).$$

Indeed, after separating the cases $xy \geq 0$ and $xy < 0$, this follows respectively from

$$|a - b|^{p-1} \leq a^{p-1} - b^{p-1} \quad (a \geq b \geq 0)$$

and

$$(a + b)^{p-1} \leq 2^{p-2}(a^{p-1} + b^{p-1}) \quad (a, b \geq 0).$$

Let a be a minimizer of $t \mapsto \mathbb{E}|U - t|^p$, and put $Z = U - a$. Then $\mathbb{E}\phi(Z) = 0$, and $\mathbb{E}|Z|^p \leq \mathbb{E}|U|^p$. If Z' is an independent copy of Z , then

$$\begin{aligned} \mathbb{E}|U - V|^p &= \mathbb{E}|Z - Z'|^p \\ &\leq 2^{p-2} \mathbb{E}[(Z - Z')(\phi(Z) - \phi(Z'))] \\ &= 2^{p-1} (\mathbb{E}|Z|^p - \mathbb{E}Z \mathbb{E}\phi(Z)) \\ &= 2^{p-1} \mathbb{E}|Z|^p \leq 2^{p-1} \mathbb{E}|U|^p. \end{aligned}$$

Sharpness for $1 < p \leq 2$ follows by taking $U_N = N^{1/p}$ with probability $1/N$ and $U_N = 0$ otherwise; then $\mathbb{E}|U_N|^p = 1$ and $\mathbb{E}|U_N - U'_N|^p = 2(1 - 1/N) \rightarrow 2$. Sharpness for $p \geq 2$ follows from $U = \pm 1$ with equal probabilities, for which $\mathbb{E}|U - U'|^p = 2^{p-1}$. $\square$

Proposition 1. *For the constants c_n of Proposition 55 in the source paper,*

$$c_n^p \leq \begin{cases} 2, & 1 < p \leq 2, \\ 2^{p-1}, & 2 \leq p < \infty. \end{cases}$$

Proof. Fix $F \in L_p(\{0, 1\}^n)$ with $\|F\|_p = 1$, and fix an admissible pair A, B . Condition on the coordinates belonging to $A \cap B$. Given those common coordinates, $T_A(F)$ and $T_B(F)$ are iid copies of the conditional distribution of F over the remaining coordinates. Lemma 1, applied conditionally and then averaged over the common coordinates, gives the stated upper bounds. $\square$

Theorem 1. *For Proposition 55 of the source paper,*

$$c = \sup_n c_n = \begin{cases} 2^{1/p}, & 1 < p \leq 2, \\ 2^{1-1/p}, & 2 \leq p < \infty. \end{cases}$$

Moreover, if $p \geq 2$, then $c_n = 2^{1-1/p}$ for every $n \geq 1$.

Proof. The upper bounds follow immediately from Proposition 1.

For $p \geq 2$, choose disjoint $A, B \in [\kappa]^n$ and let F be a Rademacher function depending on one coordinate of $\{0, 1\}^n$. Then $T_A(F)$ and $T_B(F)$ are independent Rademacher variables, so

$$\|T_A(F) - T_B(F)\|_p^p = 2^{p-1}.$$

Thus $c_n^p = 2^{p-1}$ for every n .

It remains only to prove the lower bound for c when $1 < p < 2$. Let $N = 2^n$, choose disjoint A, B , and let F take the value $a = (N/2)^{1/p}$ on one atom of $\{0, 1\}^n$, the value $-a$ on another atom, and 0 on all remaining atoms. Then $\|F\|_p = 1$. For independent copies U, V of F ,

$$\mathbb{E}|U - V|^p = \frac{2(2a)^p + 4(N-2)a^p}{N^2} = 2 - \frac{4-2^p}{N}.$$

Therefore $c_n^p \geq 2 - (4-2^p)/2^n$, and letting $n \rightarrow \infty$ gives $c^p \geq 2$. The upper bound gives $c^p \leq 2$, so $c = 2^{1/p}$. $\square$

Finite c_n for $1 < p < 2$

The proof above gives

$$2 - \frac{4-2^p}{2^n} \leq c_n^p \leq 2$$

for $1 < p < 2$. Numerical optimization strongly suggests that the lower bound is the exact value for every n , but this packet does not claim that finite extremal inequality. The unresolved finite statement is:

$$\frac{1}{N^2} \sum_{i,j=1}^N |x_i - x_j|^p \leq \left(2 - \frac{4-2^p}{N}\right) \frac{1}{N} \sum_{i=1}^N |x_i|^p$$

for $N = 2^n$, $1 < p < 2$, and all real $x_1, \dots, x_N$. Equality would be attained by two opposite nonzero entries and $N-2$ zero entries.

Verification Notes

The promoted proof is analytic. The auxiliary script `code/check_finite_atom_conjecture.py` numerically optimizes the finite $1 < p < 2$ atom problem for small N . It is included only as evidence for the remaining finite extremal route; it is not used to prove Theorem 1.

Scope and Limitations

This packet answers the exact value of c , and it answers the exact finite constants c_n for $p \geq 2$. It does not prove the exact finite formula for c_n when $1 < p < 2$, and it does not resolve whether $S_{L_p(\mu)}$ is dichotomous for $p \neq 2$.

Novelty and Search Bounds

The run indexes were searched for arXiv:2308.07668 and L_p/L_2 constants before the earlier packet, and the new packet was checked against the refreshed registry. A bounded web phrase search for the exact source sentence and nearby terms did not reveal a later paper answering this constants question. The proof uses classical negative type of the real line for $p \leq 2$ and an elementary p -center inequality for $p \geq 2$.

Human Review Recommendation

Reviewers should check Lemma 1, especially the $p \geq 2$ translation to the p -center, and then check that the source paper's admissible overlap condition really gives conditionally iid copies after conditioning on shared coordinates. The finite $1 < p < 2$ formula should be treated as a separate open extremal inequality.

References

- [1] P. Koszmider, *On Ramsey-type properties of the distance in nonseparable spheres*, arXiv:2308.07668.
- [2] I. J. Schoenberg, *Metric spaces and positive definite functions*, Transactions of the American Mathematical Society 44 (1938), 522–536.